WONG WEI MING

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EDUCATION

Nanyang Technological University (NTU), Singapore **Aug 2023 – May 2027**

- Bachelor of Engineering (Honours), Computer Engineering with Second Major in Data Analytics
- Honours (Distinction); CGPA: 4.25/5.00 (Current)
- Relevant Modules: Data Structures & Algorithms, Sensors & Digital Control, Linear Algebra, Data Science & AI

Hong Kong University of Science and Technology (HKUST), Hong Kong **Jan 2026 – May 2026**

Semester Exchange

- UG Coursework: Machine Learning, Intelligent Robot & Embodied AI, Deep Learning in Computer Vision, Embedded AI Systems
- PG Coursework: Deep 2D & 3D Visual Scene Understanding, Intro to Aerial Robotics

ACHIEVEMENTS

- **NTU President Research Scholar** (URECA) — Undergraduate Research
- **3rd Place —** Robonation RoboSub 2026 (International Level)
- **Champion** — Singapore AUV Challenge 2025 (International Level)
- **Semifinalist** (Top 10) & **Best New All-Rounders** — Robonation RoboSub 2025 (International Level)
- **Finalist** — HacX @ HTX 2025
- **Finalist —** NTU Escendo @ Garage EEE 2025 (School Level)
- **Rank 9** — Ocean Exploration Video Challenge, MATEROV 2024 (International Level)

SKILLS

- **Programming Languages:** Python, C, C++, Java
- **Robotics & Autonomy:** ROS 2, Isaac ROS, Nav2, NVBlox, MoveIt 2, ArduPilot, PX4, Behavior Trees, SLAM, Sensor Fusion (EKF/UKF), PID Control, rmw_zenoh
- **AI & Computer Vision:** PyTorch, LeRobot ($\pi_{0.5}$), Hugging Face, YOLO, OpenCV, Roboflow
- **Simulation & Visualization:** Isaac Sim, Gazebo, MuJoCo, Foxglove, Rerun, Trackio
- **Tools & Environments:** Git, Docker, Linux/Ubuntu, WSL, Pixi (Conda/PyPI Package Manager)
- **Hardware:** NVIDIA Jetson, Raspberry Pi, Holybro Pixhawk, STM32, ESP32, Intel RealSense
- **Transferable:** Technical Leadership, Systems Thinking, Project Management, Cross-Functional Collaboration

EXPERIENCE

Mecatron (NTU Maritime Robotics Team) **Aug 2024 – Present**

Advisor | Former President & Perception Lead

- **Club & Technical Leadership:** Directed a 60-member robotics club as President while managing a 10-member perception engineering team; currently serving as Advisor guiding technical governance, system architecture design, and internal administration.
- **System Architecture & Autonomous Planning:** Architected a modular ROS 2 software stack utilizing **BehaviorTree.CPP (C++)** for state machine task sequencing and **Software-in-the-Loop (SITL) simulation**, cutting physical pool debugging time by ~40%.
- **3D Computer Vision & Perception Pipeline:** Engineered an underwater perception stack combining **YOLO object detection** and **Depth Anything v3** for real-time 3D target localization and visual navigation; conducted stereo vision experiments to enhance spatial perception in low-visibility environments.
- **Curriculum & Workshop Mentorship:** Designed and delivered technical training covering Computer Vision (CNNs, YOLO model deployment) and ROS 2 probation workshops, training 150+ NTU students and onboarding 10+ core perception engineers.
- **Competition Track Record:** Software and perception contributions directly powered the AUV to achieve **Champion at SAUVC 2025**, **Top 10 Semifinalist & Best New All-Rounders at RoboSub 2025**, and **3rd Place at RoboSub 2026 (International)**.

Cheng Kar-Shun Robotics Institute (CKSRI), HKUST **Mar 2026 – May 2026**

Undergraduate Research Assistant | Supervised by Prof. Ling Shi (Director, CKSRI)

- **Mobile Manipulator System Integration:** Executed full-stack hardware-software bring-up for an **AgileX Scout Mini 4WD UGV** mounted with an **AgileX Piper 6-DOF arm**, configuring `ugv_sdk` C++ CAN bus communication and `scout_ros2` node lifecycles.
- **IMU-Free Laser Odometry & SLAM Navigation:** Integrated **Richbeam 2D LiDAR** (`Lakibeam_ROS2_Driver`) with **`rf2o_laser_odometry`** to compute real-time range-flow 2D base odometry on an IMU-less robot; combined with **Nav2** and **reflective marker intensity tracking** for sub-centimeter base positioning.
- **MoveIt 2 Trajectory Planning & Public Showcase:** Engineered a **MoveIt 2** motion planning pipeline on the Piper arm for interactive chess playing, deploying the system live at the **2nd HKUST AI Film Festival** achieving **<1mm positioning error** across **>8 hours of continuous live operation**.

SkyrisAI **Dec 2025 – Jan 2026**

ROS Architecture Engineer

- **ROS 2 System Architecture & Aerostack Integration:** Architected a modular ROS 2 software stack for an autonomous flying companion robot, experimenting with the **Aerostack2** framework to unify low-level flight control with high-level mission orchestration.
- **Visual-Inertial SLAM & VIO Evaluation:** Benchmarked and evaluated Visual-Inertial Odometry (VIO) and SLAM pipelines (including **VINS-Fusion**) in Gazebo SITL simulation to optimize 3D localization accuracy under dynamic flight profiles.
- **Behavior Tree & Autonomous Owner-Following:** Programmed custom **Behavior Tree** action nodes and state machine logic to enable real-time owner-following behavior and multi-layered safety fail-safes for aerial companion interaction.

Home Team Science & Technology (HTX) **May 2025 – Dec 2025**
Artificial Intelligence & Robotics Engineer Intern

- **Jetson–Pixhawk "Common Core" System Architecture:** Engineered a modular, multi-domain autonomy stack pairing NVIDIA Jetson (high-level AI) with Pixhawk 6C Mini (ArduPilot Rover via FastDDS / micro-ROS bridge), creating a reusable hardware-software foundation across UGV, UAV, and AUV platforms.
- **NVIDIA Isaac ROS NVBlox & Nav2 Integration:** Deployed CUDA-accelerated Isaac ROS **NVBlox** (3D mapping at 4.9 FPS, 5cm voxel size, 63% GPU load) on Jetson Orin with Intel RealSense D435i, streaming real-time 3D costmaps into **Nav2** for vision-based obstacle avoidance.
- **System Bottleneck Profiling & Telemetry:** Profiled real-time DDS queue saturation, serialization latency, and CUDA memory contention using **Foxglove Studio** and `jetson-stats`, establishing hardware requirements for AGX Orin migration.
- **Agentic AI Integration & Task Planning:** Integrated NASA JPL **ROSA (ROS Agent)** for natural-language cognitive task planning and dynamic ROS tool execution with <2s command dispatch latency.

PROJECTS & RESEARCH

Scene Graph-Grounded Semantic World Model Planning for Mobile Manipulation **Aug 2026 – Present**
NTU Final Year Project (FYP) | Supervised by Prof. Yoonchang Sung (NTU) & Dr. Dikai Liu (NVIDIA)

- **System Architecture & Research Focus:** Conducting literature review and architecture formulation on Graph World Models, extending 3D functional scene graphs to incorporate closed-loop physical feasibility verification for mobile manipulation.
- **Technical Methodology:** Investigating Vision-Language Models (VLMs) to evaluate base-stance positioning offsets ($\Delta x, \Delta y$) and kinematic reachability constraints prior to motor execution to prevent action failure.

VLASH-Piper: Real-Time Async VLA Deployment on Edge | GitHub **Feb 2026 – May 2026**
HKUST COMP4901D (Embedded AI Systems) | Best Project Runner-Up Award

- Fine-tuned $\pi_{0.5}$ Vision-Language-Action (VLA) policy and integrated Hugging Face LeRobot (v0.4.1) for AgileX PIPER 6-DOF arm control over CAN bus.
- Deployed on NVIDIA Jetson AGX Orin using multi-threaded async inference (VLASH framework) to decouple camera encoding from model execution, achieving a **30Hz real-time control loop** (<35ms latency) for zero-shot tabletop manipulation.

Underwater DA3: LoRA Fine-Tuning for AUV Depth Estimation | GitHub **Feb 2026 – May 2026**
HKUST COMP4471 (Deep Learning in Computer Vision)

- Adapted Depth Anything 3 (DA3) Mono Metric Large model for underwater environments by injecting rank-8 LoRA into DINOv2-L ViT backbone to overcome light attenuation and backscatter.
- Trained on NVIDIA A100 GPUs using SILog + Sobel gradient loss, reducing **AbsRel error by 59%** and boosting $\delta < 1.25$ accuracy from **0.94% to 56.8%** on MIMIR-UW SeaFloor benchmark.
- Fused per-pixel metric depth with YOLOv11-seg instance masks and Kalman filter target tracking within a ROS 2 Jazzy AUV 3D localization stack.

Scaling LLaVA-3D for Large-Scale 3D Scene Understanding | GitHub **Feb 2026 – May 2026**
HKUST COMP5422 (3D Visual Scene Understanding) | Supervised by Prof. Dan Xu

- Extended LLaVA-3D with plug-in compression strategies (attention-score pruning, spatial K-Means clustering) to improve 3D Large Multimodal Model scalability without retraining base model.
- Spatial K-Means clustering ($K = 512$) reduced token count from **~3,096 to 512** (~6x compression) while boosting ScanQA validation EM@1 accuracy from **46.2% to 48.8%**.

Visual-Language Behavior Planning for Autonomous Underwater Vehicles **Aug 2024 – May 2025**
NTU URECA Research Project (CCDS24070) | Supervised by Assoc Prof Nicholas Vun | Awarded NTU President Research Scholar

- Deployed 4-bit AWQ quantized **VILA Vision-Language Model (VLM)** on **NVIDIA Jetson Orin NX** (16GB, Jetson AI Lab Docker container) operating under a 25W power budget for swimming pool AUV operations (~3.1s latency).
- Integrated interactive Visual Question Answering (VQA) with **NanoDB Vector Database** to query behavior components and assemble XML **Behavior Trees (BT)** executed via **BehaviorTree.CPP** in ROS 2.
- Validated end-to-end pipeline in SITL simulation and physical swimming pool trials on real AUV for basic autonomous actions (seeing obstacles, turning to avoid obstacles, and basic movement).